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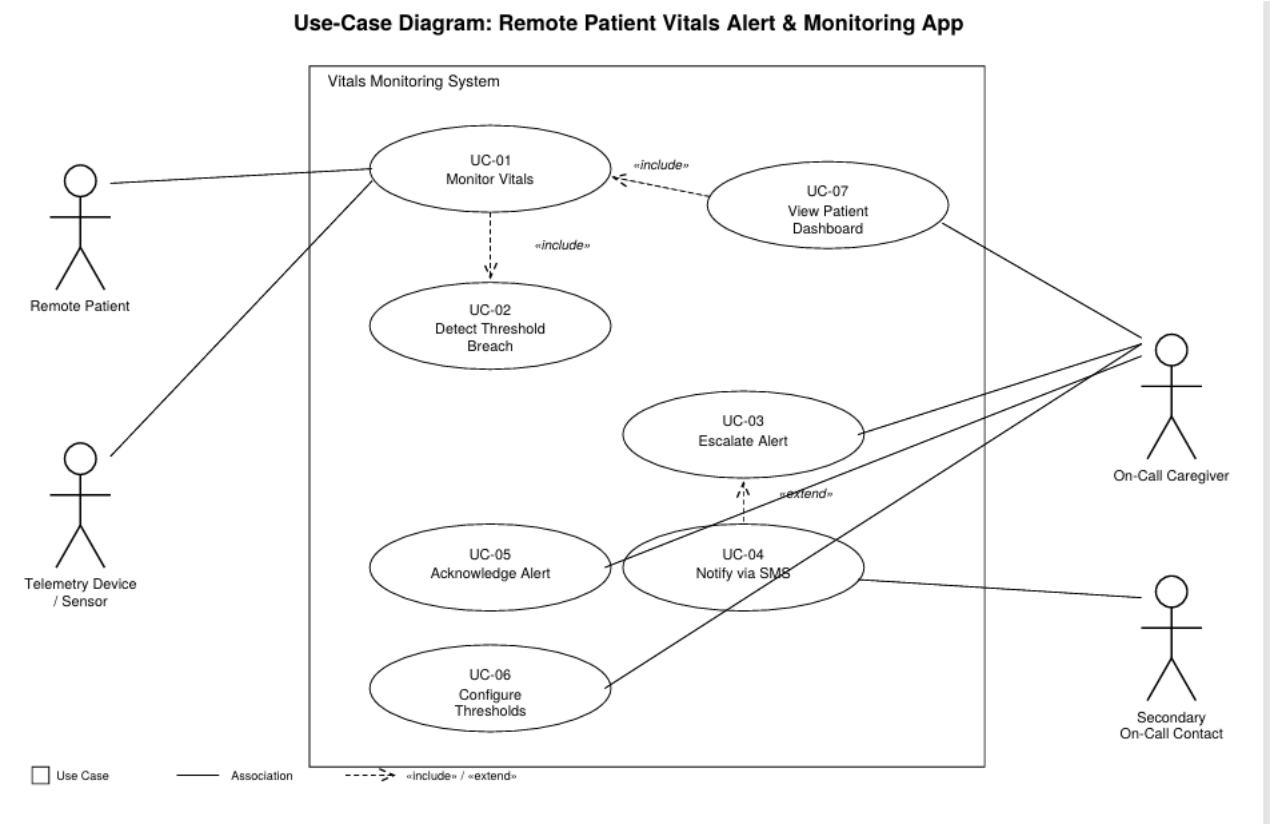
Probem Statement 16 : Patient Vitals Alert & Monitoring App

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall continuously evaluate ingested patient vitals against configured clinical thresholds and flag critical anomalies within 2 seconds.	High	Pass: A heart rate reading above 140 BPM triggers an immediate escalation alert. Fail: The metric spike is ignored or the alert is delayed more than 5 seconds.	Given requirement — core safety function of the monitoring pipeline
FR-002	Functional	The system shall allow an on-call caregiver to configure patient-specific vital sign thresholds (e.g., SpO2 lower limit, heart rate range, blood pressure range).	High	Pass: When a caregiver sets a custom SpO2 threshold of 90%, the system flags any subsequent reading below 90% for that patient. Fail: The system applies the default threshold instead of the configured one.	Post-operative patients have individualized clinical baselines, so a single fixed threshold is unsafe.
FR-003	Functional	The system shall escalate a critical alert to the on-call caregiver via push notification, and via SMS if the alert is not acknowledged within 60 seconds.	High	Pass: An unacknowledged push notification results in an SMS being sent to the caregiver at the 60-second mark. Fail: No SMS is sent within 90 seconds of the original alert.	A missed push notification must not result in a missed emergency; escalation guarantees the alert reaches the caregiver.
FR-004	Functional	The system shall allow the on-call caregiver to acknowledge an active alert and record a response action against it.	Medium	Pass: Tapping "Acknowledge" timestamps the response, records the caregiver's ID, and updates the alert status to "Handled." Fail: The alert status remains "Open" after acknowledgment.	An auditable response log is required for clinical accountability and post-incident review.
FR-005	Functional	The system shall display a real-time dashboard of a patient's current vitals (SpO2, Heart Rate, BP) to the assigned on-call caregiver.	Medium	Pass: The dashboard reflects a newly ingested vital reading within 2 seconds of receipt. Fail: A displayed value is more than 5 seconds stale relative to the latest telemetry	Caregivers need current, at-a-glance vitals to make timely clinical decisions, not just alert notifications.
NFR-001	Nonfunctional (Performance & Security)	The telemetry ingestion gateway shall support at least 500 concurrent continuous telemetry streams with 99.99% uptime	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load. Fail: Latency or uptime targets are missed under the same test conditions.	Given requirement — the system must scale to a hospital-wide patient population without service degradation.

NFR-002	Nonfunctional (Security & Compliance)	The system shall encrypt all patient vitals data in transit (TLS 1.2 or higher) and at rest (AES-256), in line with HIPAA data-protection requirements. Functional	High	Pass: A penetration/security test confirms TLS 1.2+ is enforced on all data-in-transit channels and AES-256 encryption is applied to stored data. Fail: Any channel or stored record is found unencrypted or using a weaker cipher.	Patient health data is highly sensitive and regulated; a breach carries legal and safety consequences.

Use -Case Diagram



Use-Case Flow Specification

Use Case: Escalate Alert — Remote Patient Vitals Alert & Monitoring App

Primary Actor:

On-Call Caregiver

Secondary Actors:

Remote Patient (source of vitals), Telemetry Device/Sensor

Preconditions:

• The patient's vitals-monitoring session is active and telemetry is streaming.

• Clinical thresholds have been configured for the patient (see FR-002).

• The on-call caregiver is registered and reachable via push notification and SMS.

Postconditions:

• The alert is logged with a final status of either "Acknowledged" or "Escalated – Secondary Contact Notified."

• An auditable timestamped record of the alert and the caregiver's response exists in the system.

Main Success Scenario:

1. The system continuously evaluates incoming patient vitals (SpO2, Heart Rate, BP) against the configured clinical thresholds.

2. The system detects that a vital reading has breached its configured threshold (e.g., Heart Rate > 140 BPM).

3. The system generates a critical alert and flags it within 2 seconds of the breach.

4. The system sends a push notification of the alert to the on-call caregiver's device.

5. The caregiver reviews the alert details, including the patient's current vitals dashboard.

6. The caregiver taps "Acknowledge" within 60 seconds of receiving the notification.

7. The system timestamps the acknowledgment, records the caregiver's ID, and updates the alert status to "Handled."

8. Use case ends successfully.

Alternate Flow — Alert Not Acknowledged in Time:

• 1. 60 seconds elapse after the push notification with no acknowledgment from the caregiver.

• 2. The system automatically escalates by sending an SMS alert to the same caregiver and notifying a secondary on-call contact.

• 3. If the secondary contact acknowledges, the system logs the alert status as "Escalated – Secondary Contact Notified" and the use case ends.

• 4. If no acknowledgment is received from either contact within an additional 2 minutes, the system escalates to the hospital's emergency response protocol and logs the alert as "Escalated – Emergency Protocol Triggered."